

2. *Shape of the Eigenfunction $v_{s,E}$* : To precisely understand how $\lambda_{s,E}$ varies in E , we must not only understand the operator $F_{s,E}$ but also the associated eigenfunction $v_{s,E}$. This differs from prior work, such as [Bap14], which circumvented the analysis of $v_{s,E}$ by instead estimating the action of $F_{s,E}$ on certain test vectors (which loses careful control on $\lambda_{s,E}$ for $E \approx E_0$). We approximate the eigenfunction $v_{s,E}$ in Section 8, where we begin by defining an explicit function $\tilde{u}$ by

$$\tilde{u}(x) = (t^2 \ln K)^{-1} \cdot \mathbb{1}\{|x| \leq Ct^2\} + |x|^{-(2-s)} \cdot \mathbb{1}\{Ct^2 \leq |x|\}. \quad (1.10)$$

Observe that, if s is very close to 1, the function $\tilde{u}(x)$ behaves as $(t^2 \ln K)^{-1}$ for $x \leq Ct^2$ and as x^{-1} for $x \geq Ct^2$. This is the general behavior we expect for the eigenfunction $v_{s,E}$. However, the precise location and nature of the crossover around $x = O(t^2)$, at which $v_{s,E}(x)$ changes from behaving as $x^{-1} \sim t^{-2}$ to as $(t^2 \ln K)^{-1}$, is not transparent to us. So, instead of comparing $v_{s,E}$ directly to $\tilde{u}$, we will compare $F_{s,E}v_{s,E}$ to $F_{s,E}\tilde{u}$. Specifically, as Lemma 8.5, we show that

$$c(F_E \tilde{u})(x) \leq (F_{s,E}v_{s,E})(x) \leq C(F_E \tilde{u})(x), \quad (1.11)$$

where the constants c and C are uniform in K . While useful, this does not explain how $v_{s,E}$ varies with E and, indeed, we do not know how to do this.

We bypass this by making use of the identity

$$\ln \lambda_{s,E} = \lim_{n \rightarrow \infty} \frac{\ln \|F_E^n \tilde{u}\|_1}{n}. \quad (1.12)$$

The benefit of this representation is that it replaces $v_{s,E}$ with the function $\tilde{u}$ that is independent of E . However, it comes at the expense of having to compute high powers of $F_{s,E}$ when applied to $\tilde{u}$.

3. *Estimating Iterates of $F_{s,E}$* : To deduce the monotonicity of $\lambda_{s,E}$, we analyze the difference $\ln \lambda_{s,E_2} - \ln \lambda_{s,E_1}$ using (1.12). We have

$$\|F_{s,E_2}^n \tilde{u}\|_1 - \|F_{s,E_1}^n \tilde{u}\|_1 = \int_{-\infty}^{\infty} ((F_{s,E_2}^n - F_{s,E_1}^n) \tilde{u})(x) dx. \quad (1.13)$$

We note the identity

$$F_{s,E_2}^n - F_{s,E_1}^n = \sum_{j=0}^{n-1} F_{s,E_2}^{n-j-1} (F^\circ) F_{s,E_1}^j, \quad (1.14)$$

where we define

$$(F^\circ u)(x) = (F_{s,E_2} - F_{s,E_1})u(x) = \frac{t^{2-s}}{|x|^{2-s}} \int_{-\infty}^{\infty} (\rho_{s,E_2}(-y - t^2 x^{-1}) - \rho_{s,E_1}(-y - t^2 x^{-1})) u(y) dy.$$

When $|x| \geq Ct^2$ and y is close to 0 (which is where the above integral will mainly be supported), (1.9) implies that $\rho_{s,E_2}(-y - t^2/x) - \rho_{s,E_1}(-y - t^2/x) < c(E_2 - E_1)$. This suggests that F° should act as a negative operator for functions supported outside of $[-Ct^2, Ct^2]$.

In Section 9, we study F° by justifying this reasoning. In particular, as Lemma 9.3, we confirm that $F^\circ F^j \tilde{u}(x)$ is negative for $|x| \geq Ct^2$ and bound it away from zero. The contribution from $|x| \leq Ct^2$ might be positive, so we also provide a bound on this quantity in the same Lemma 9.3, using (1.8). Throughout these analyses, we make repeated use of (1.11), to bound high powers of $F_{s,E}$ applied to $\tilde{u}$ by a multiple of $v_{s,E}$ (on which $F_{s,E}$ acts diagonally).

In Section 10, we use the estimates from Lemma 9.3 as inputs to show that the negative contributions from the regime $|x| \geq Ct^2$ outweigh the positive ones from when $|x| \leq Ct^2$. We therefore deduce that the $F_{s,E_2}^{n-j-1} (F^\circ) F_{s,E_1}^j$ from (1.14) essentially act as nonpositive operators on $\tilde{u}$, when $E_1 < E_2$ are close to E_0 . Combining (1.13) and (1.14), we find $\|F_{s,E_2}^n\|_1 - \|F_{s,E_1}^n\|_1 \leq 0$. Then (1.12) shows that $\lambda_{s,E}$ is weakly decreasing near E_0 , which is stated as Lemma 10.2. This argument